

application concerned a game apparatus the board stated that "game rules" form part of "the regulatory framework agreed between [or with] players concerning conduct, conventions and conditions that are meaningful only in a gaming context. They govern the conduct and actions of the players during game play. Disregarding any effects and advantages inherent in the game rules themselves, the board was unable to identify any further technical effects in the particular manner of technical implementation that might have rendered it non-obvious to the skilled person.

In T 1225/10 the board also adopted the approach set out in T 1543/06 and held that it was decisive how excluded subject-matter was technically implemented, and whether that implementation was obvious. Such a consideration focused on any further technical effects of the implementation of the excluded subject-matter over and above those inherent in the excluded subject-matter itself (T 1543/06, T 1782/09). In T 1225/10 the board came to the conclusion that using points on the player character that overlapped with a target object was a computationally effective and efficient approach to determine the collision side on a target object for a variety of such objects. That effect was not the inevitable result of programming the game rules per se. It was rather the direct technical consequence of the particular technical way selected data was used to determine a display state. Therefore the technical effect was affirmed.

9.2.15 Mental acts / modelling

In T 914/02 the board held that the involvement of technical considerations is not sufficient for a method which may exclusively be carried out mentally to have technical character. Technical character may be provided through the technical implementation of the method, resulting in the method providing a tangible, technical effect, such as the provision of a physical entity as the resulting product or a non-abstract activity, such as through the use of technical means. The board rejected a claim directed to an invention involving technical considerations and encompassing technical embodiments on the grounds that the invention as claimed could also be exclusively performed by purely mental acts excluded from patentability under Art. 52(2)(c) EPC (see T 619/02, OJ 2007, 63; T 388/04, OJ 2007, 16, T 1073/06).

In T 930/05 the application was concerned with a method for modelling a process network. The board held that the method lacked technical character because though the claimed subject-matter comprised technical embodiments, implementations that could be deemed non-technical were also conceivable.

In T 1171/06 the application related to object-oriented modelling, namely a method and a device for modelling a mechatronic system in a motor vehicle. The board held that a model used in software development was not imparted a technical effect because it served documentation or communication, even if its subject-matter was a technical system. The board continued the approach established in T 354/07. In that case, which concerned the automatic generation of program code from a model, the board had noted that conceptual processes and meta methods for software production generally had no technical features relevant for patentability and thus could not provide a basis for inventive step unless it

could be demonstrated, in a given case, that there was a direct causal connection with a technical effect relevant for solving a technical problem.

9.2.16 Logistics

In T. 696/06 the application concerned a data processing system intended to facilitate confronting an offer and a demand in the field of transporting travellers or goods. Essentially the system executed business processes and transactions, like mediating offers and demands, which were typical for a transportation broker or freight exchange service and which the board held not to be technical.

In T. 912/05 the application related to a mail delivery system that could deliver by physical and/or electronic means. The board held that methods of mail delivery can qualify as inventions if they involve "technical means", with mail delivery by e-mail qualifying as such.

9.3. Combination invention

9.3.1 Existence of a combination invention

In assessing the inventive step involved in an invention based on a combination of features, consideration must be given to whether or not the state of the art suggested to a skilled person precisely the combination of features claimed. The fact that an individual feature or a number of features were known individually does not conclusively show the obviousness of a combination (T. 37/85, OJ 1988, 86; T. 656/93; T. 666/93; T. 1018/96). The question is not whether the skilled person, with access to the entire prior art, **could** have made the combination according to the invention, but whether he actually **would** have done so in expectation of an improvement (T. 2/83, OJ 1984, 265; T. 713/93; T. 223/94; T. 406/98). When assessing inventive step in a combination invention, the decisive criterion is not whether individual elements of the combination were known and obvious from prior art, but whether the state of the art would lead a skilled person to this particular overall combination of (possibly already known) features. Were this not so, it would be impossible for a combination consisting exclusively of known individual features to involve an inventive step (T. 37/85, OJ 1988, 86; T. 388/89; T. 717/90; T. 869/96).

A mere aggregation of features must be distinguished from a combination invention.

The existence of a combination invention requires that the relationship between the features or groups of features be one of **functional reciprocity** or that they show a combinative effect beyond the sum of their individual effects. In T. 1054/05 the board stated that two features interacted **synergistically** if their functions were interrelated and lead to an additional effect that went beyond the sum of the effects of each feature taken in isolation. It was not enough that the features solved the same technical problem or that their effects were of the same kind and added up to an increased but otherwise unchanged effect (see also T. 926/11, in which it was found there was no synergistic effect between the feature groups). See also in this chapter I.D.9.2.2.

In T 406/98 the board found that as a rule, particularly when large numbers of citations were involved, it was necessary to ask **why** the skilled person would consider documents in that specific combination, and whether, **not knowing the invention**, he had reason to do so. In this case, a complete solution to the problem required deliberate selection from a large number of citations.

In T 55/93 the appellant's argument that the alleged invention should have been regarded as a mere aggregation of solutions of two independent partial problems which were not interrelated, was not accepted by the board. This reasoning could only stand if the primary and more general problem were already known and solved by the prior art. In the case in point, not only could the primary problem underlying the contested patent neither be found nor be derived from the prior art documents, but also the claimed features complemented each other. The board stated that the features were functionally linked together, which was the very characteristic of a combination invention. The board stated it was no consequence whether, in a combination invention, possibly all features were already known per se, separately (T 37/85, OJ 1988, 86). It was wrong to select on the basis of a plurality of partial problems to be solved, the respective constructional means used in the apparatus combination, or the steps of the method worded in terms of functional features, which by working together provided a solution to the problem taken as a whole. The non-obviousness of a combination claim turned on the simultaneous application of all its features (T 175/84, OJ 1989, 71). A combination effect was also acknowledged in T 120/88, T 731/94, T 434/95, T 897/95, T 1201/13.

9.3.2 Partial problems

In patent law terms, the existence of a combination of features, i.e. of a combination invention, is to be viewed differently from the mere existence of partial problems, i.e. of an aggregation of features. According to current case law, partial problems exist if the features or sets of features of a claim are a mere aggregation of these features or sets of features (juxtaposition or collocation) which are not **functionally interdependent**, i.e. do not mutually influence each other to achieve a technical success over and above the sum of their respective individual effects, in contrast to what is assumed in the case of a combination of features. What has to be established is whether each set of features is separately obvious in the light of the prior art (T 389/86, OJ 1988, 87; T 387/87; T 294/90; T 363/94; T 926/11; T 1587/14). It should also be borne in mind that solutions to partial problems in differing technical fields must be assessed on the basis of the knowledge and expertise of the person skilled in the art where the solution is found (T 32/81, OJ 1982, 225; T 324/94).

In T 389/86 (OJ 1988, 87) the relationship between the two groups of features was not one of functional reciprocity. The board ruled that in such circumstances no combinative effect could be advanced in support of inventive step; rather the question was whether each group, taken singly, was obviously derivable from the prior art. For the subject-matter of the claim to be inventive, it sufficed if one of these groups was (see also T 345/90, T 701/91, T 94/05, T 450/06, T 102/08, T 619/08, T 2097/10).

In T.1836/11, the patent concerned a two stage turbocharger for an internal combustion engine. The respondent (patent proprietor) argued that the backsweep and dual feed worm features could not be considered separately since they synergistically contributed to the stated aim of increasing efficiency. The board did not see any synergy between the two features. That the features both served the same overall purpose did not establish, in the board's view, a functional reciprocity between the two.

In T.130/89 (OJ 1991, 514) the technical problem intended to be solved by the claimed invention consisted of two technically independent partial problems, each solved independently by one of the claimed subject-matter's features. The board held that the independence of the claimed subject-matter's features (each producing a different effect) meant that in assessing inventive step the two closest states of the art had to be considered to enable each of the two partial problems to be defined. It concluded that since each of the partial problems was solved by means which merely performed their known functions, each partial solution was obvious and the invention thus lacked inventiveness. In T.597/93 the board again saw no inventive step in combining the claim's two features – both known per se – since they related to the solving of two entirely separate partial problems. It cited T.687/94 which held that in such cases the solutions could be assessed separately against the prior art (see also T.315/88, T.65/90, T.2110/08).

In T.711/96 the board found that characterising features (a) and (b) functioned completely independently of each other; there was no functional interplay (combination) between them. Although the setting for one value (e.g. spread) could indirectly affect that for the other (e.g. quantity), in that spread and quantity could both be adjusted upwards to maintain constant distribution, the two features were not directly related. In other words, the characterising features did not necessarily influence each other, although they could do. The board therefore assessed the inventive step of the two features separately, and concluded that both partial problems were obvious (see also T.1585/07).

In T.410/91 the board of appeal stated that no inventive step was involved since, although all the measures in claim 1 contributed to an increase in the efficiency of the plant, that contribution was based on known, different individual effects which resulted in these measures being executed in a manner expected by the skilled person. The subject-matter of claim 1 therefore involved the stringing-together of known measures which displayed their characteristic effects; no **synergistic** effect based on a combination of the individual measures was discernible in the sense of a mutual influence on their respective operation (see also T.144/85, T.141/87, T.407/91, T.1277/01).

In T.204/06 the board recalled that the "could-would approach" involved asking whether the skilled person would have – as opposed to could have – taken a certain step towards the invention in expectation of some improvement or advantage (T.2/83, OJ 1984, 265). This approach should not be taken to mean that inventions involving known design choices were non-obvious if only the number of choices was sufficiently great. It did imply, however, that if the skilled person expects some advantage of each feature in a claim and obtains no more than this advantage, then the claimed feature combination is obvious. It follows that any combination of features having known advantages (and disadvantages) is obvious unless it provides an unexpected effect (see also T.2044/09).

In T.547/17 the board agreed that the two distinguishing features of the claim at issue contributed to maximising the output current, however, this did not mean that the resulting output current was more than the sum of the individual contributions of the two ratios. The mere fact that both features contributed to maximising the same physical quantity (here the output current) was not sufficient to conclude that there was synergy between them. The board further agreed with the respondent that the two ratios according to the distinguishing features were interrelated. However, whether there was synergy between two features was a matter of whether their technical effects produced a combined technical effect which was greater than the sum of their individual technical effects, and not simply a matter of geometrical interrelation.

9.4. Combination of teachings

In T.1014/07 the examining division considered the subject-matter of claim 1 as obvious for the reason that each of the claimed features had been disclosed in the prior art. However, the board stated that the mere existence of teachings in the prior art is not a conclusive reason for explaining that the skilled person would have combined these teachings in order to solve the problem that he or she is confronted with. For the determination of the obviousness or non-obviousness of claimed subject-matter, it is not decisive that teachings are known – it must be decided whether or not the skilled person would have combined the known teachings such as to arrive at the claimed subject-matter when attempting to solve the underlying technical problem. Thus, the combination of known teachings may result in non-obvious subject-matter, namely when the skilled person is not motivated, for example by promptings in the prior art, to make such a combination. Under these circumstances the presence of any special effect arising from the combination is not necessary to establish an inventive step (see also T.1861/16).

9.5. Technical disclosure in a prior art document

In line with the established case law of the boards of appeal, when investigating inventive step it should be borne in mind that the technical disclosure in a prior art document should be considered in its entirety, as it would be done by a person skilled in the art and that it is not justified arbitrarily to isolate parts of such document from their context in order to derive from them technical information which would be distinct from the integral teaching of the document (T.56/87, OJ 1990, 188; T.768/90; T.223/94; T.115/96; T.717/96; T.414/98). According to T.95/90, **different parts of text in a document** can be combined if there is nothing to stop the skilled person from doing so. Any ex post facto analysis of a document, i.e. any attempt to misinterpret the disclosure of the prior art so as to distort the proper technical teaching of the disclosure in order to arrive at the claimed subject-matter, should be avoided since this would conceal the real technical contribution of the invention (T.1967/08).

9.6. Features not contributing to the solution of the problem

According to the established case law of the boards of appeal, features which do not contribute to the solution of the problem set in the description are not to be considered in assessing the inventive step of a combination of features (T.574/92). In